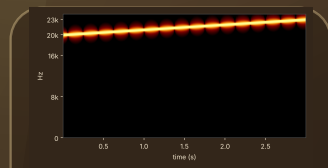
A
SAFER
WALK

SMART CANINE REPELLER

AN AI THAT HEARS A BARK
AND ANSWERS WITH A SOUND
ONLY DOGS CAN HEAR

EXPLORE >





20-23 KHZ

THE PROBLEM

WHY EXISTING SOLUTIONS FALL SHORT



Real risk

Stray dog attacks threaten walkers, cyclists, children and delivery workers.



Too late

Manual repellents and motion sensors act only after the danger is already close.



False alarms

Simple "loud sound" triggers fire on traffic, horns and music.

We need a system that knows what the sound is, not just how loud it is.

OUR SOLUTION

A MICROPHONE LISTENS, AI DECIDES "DOG OR NOT", A SPEAKER ANSWERS

Hands-free

Runs without the user doing anything.

Smart

Checks what the sound is, not how loud.

Humane

No physical contact, no chemicals.

Low cost

Just a phone or laptop with mic and speaker.

Early warning

A bark often comes before an attack.

Personalised

Add your own dog's data so it is never repelled.

HOW IT WORKS

SEVEN STEPS FROM SOUND TO RESPONSE

01 **Listen**
Mic records continuously

02 **Windows**
1 s pieces, 50% overlap

03 **MFCC**
40 x 44 feature grid

04 **CNN**
Bark probability 0 to 1

05 **YAMNet**
Checks it is a dog; rejects music and speech

06 **Confirm**
Enough windows pass both checks; own dog skipped

07 **Repel**
20-23 kHz, 3 seconds

ILLUSTRATIVE BARK WAVEFORM



The repellent fires only when both the CNN and YAMNet agree it is a dog.

EVALUATION

MEASURED ON TEST CLIPS THE MODEL NEVER TRAINED ON

ACCURACY

0.702



PRECISION

0.675



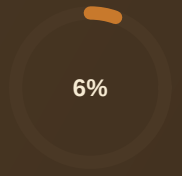
RECALL

0.029



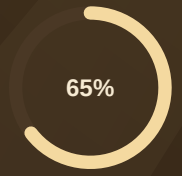
F1 SCORE

0.055

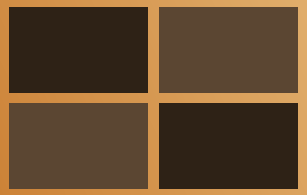


ROC-AUC

0.646



CONFUSION MATRIX



See Cell 8 output

Reading: accuracy and precision are moderate, but recall is very low, so at the 0.5 threshold the CNN misses most barks. More dog recordings and a lower threshold are needed.

DATASET

TWO CLASSES UPLOADED BY THE USER

05
06
07



Class 1: Dog barking

Option 1 upload. Barks are augmented 4 ways to cover a small number of recordings.



Class 2: Other sounds

Option 2 upload: music, car horn, siren, speech, wind, rain and more.

SPLIT BY FILE, SO NO CLIP LEAKS BETWEEN SETS

TRAIN 70% | VALIDATION 15% | TEST 15%

1-SECOND WINDOWS
SILENT PARTS REMOVED
NOISE
REAL BACKGROUND MIX
PITCH +/-2 SEMITONES
TIME SHIFT
MY DOG CLIPS (OPTIONAL)

WHY TWO MODELS

MUSIC WAS FOOLING THE FIRST VERSION

CNN only (v1)

Trained on few dog clips and no music, so songs looked like barks.

Result: constant false alarms.

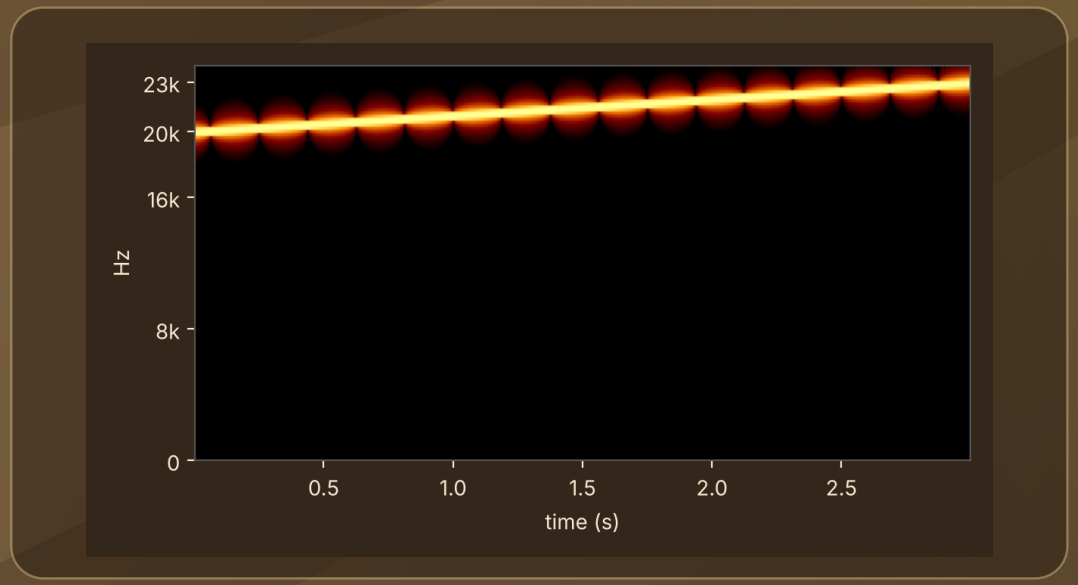
CNN + YAMNet (final)

YAMNet, trained by Google on millions of sounds, confirms "dog" and rejects music, speech and horns.

Triggers only when both agree.

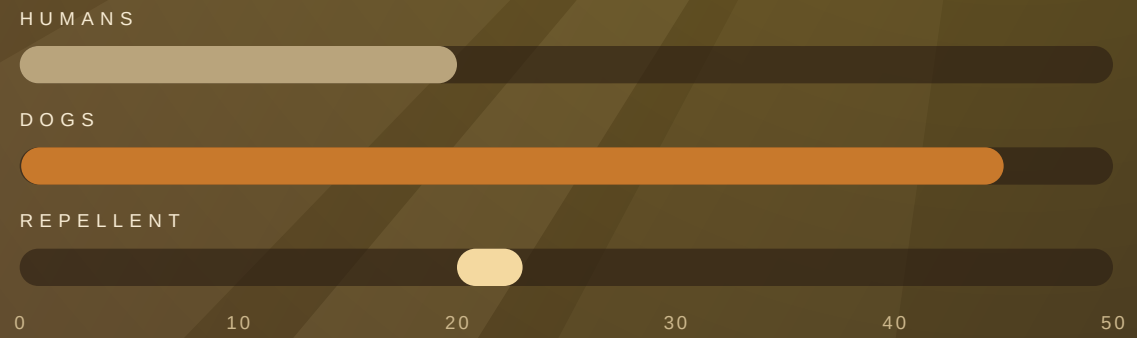
THE ULTRASONIC SOUND

SMOOTH 20-23 KHZ SWEEP FOR 3 SECONDS



REAL SPECTROGRAM OF THE GENERATED SWEEP

WHO CAN HEAR WHAT (KHZ)



- Above human hearing, so people nearby are not disturbed.
- Output runs at 48 kHz, so it cannot go above 24 kHz.
- A test-tone checkbox plays an audible sound to check the speaker.
- Limit: most phone speakers are weak above about 18 kHz.

Research prototype: ultrasonic deterrents have mixed evidence, so this is not guaranteed protection.

INTERFACE

GRADIO WEB APP WITH FOUR TABS, MOCHA AND CHAMPAGNE THEME



Analyze Audio

Upload or record, set thresholds, press Analyze & Trigger.

Live Guard

Continuous listening that triggers automatically.

My Dog

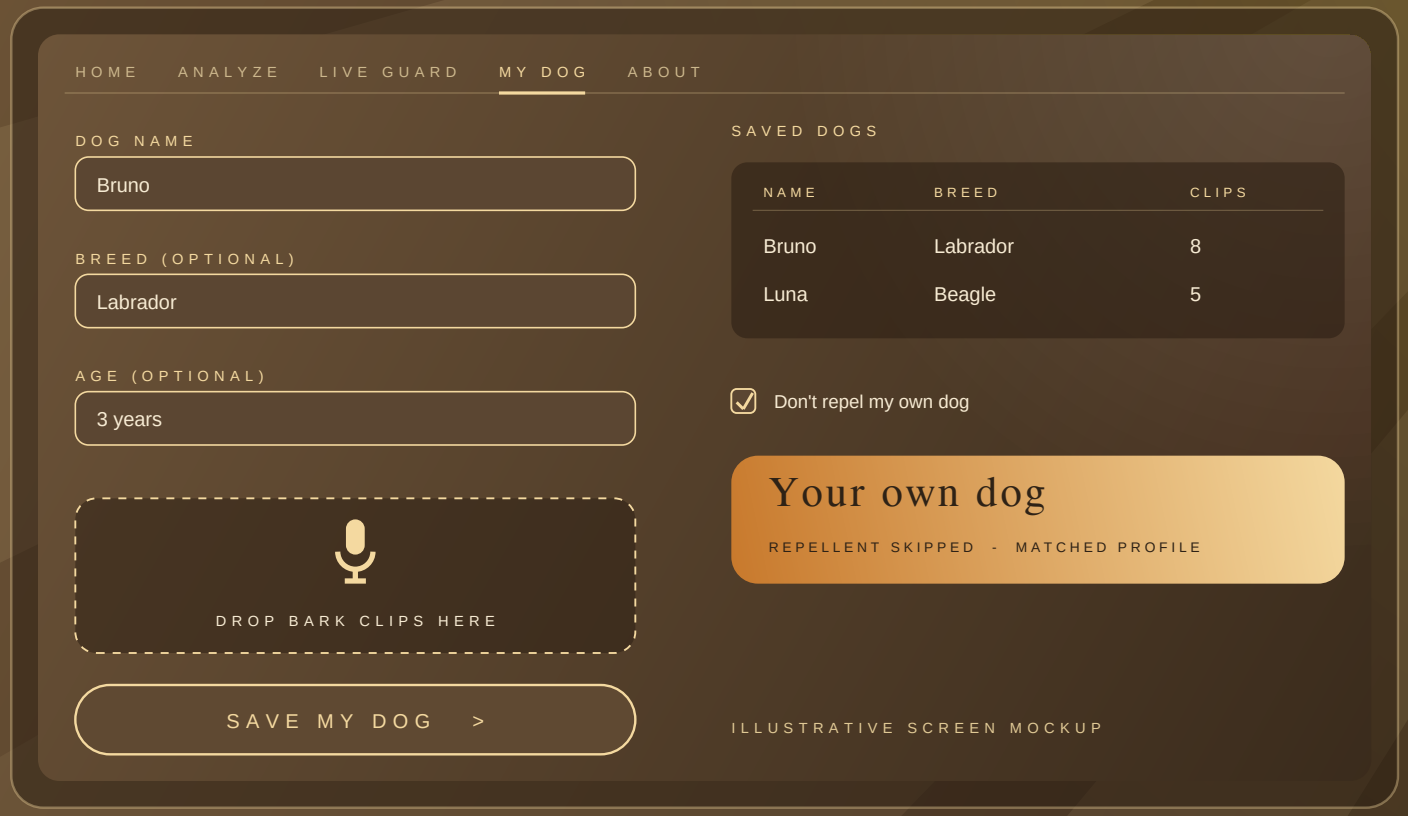
Add your dog's name, breed, age and bark clips.

About

Short explanation and safety note.

MY DOG

ADD YOUR OWN DOG'S DATA TO THE INTERFACE



01

Add

Enter name, breed and age, then upload clips of your dog barking.

02

Learn

YAMNet turns the clips into a voice fingerprint of your dog. Clips are also saved with the training data.

03

Protect

Tick "Don't repel my own dog". Matching barks are skipped. Re-run training to teach the CNN too.

Match is approximate: a very similar-sounding dog may also be skipped.

LIMITATIONS AND FUTURE WORK

HONEST LIMITS AND WHERE WE GO NEXT

Limitations

- Small dog dataset; more recordings improve accuracy.
- Phone speakers and mics are weak near 20 kHz.
- A web page cannot run in the phone background.
- Ultrasonic repellents have mixed evidence.
- Own-dog match is approximate; look-alike dogs may be skipped.

Future work

- Android app with background service using the TFLite model.
- More recordings: UrbanSound8K and field audio.
- Tune thresholds on real street audio.
- Bark-intensity scoring: warning vs aggressive barks.



THANK YOU

QUESTIONS?